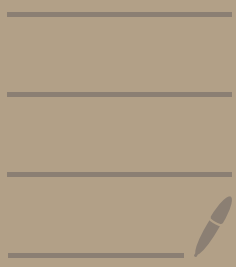
Policy Gradient Methods & Applications in LLM Post-Training

1. REINFORCE
2. Baseline & Actor-Critic
3. GAE
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Policy Gradient

Algorithms: REINFORCE (vanilla PG), Actor-Critic, NPG, TRPO, PPO, GRPO, DAPO, DSPO, ...

Setup: MDP: $(S, A, P, R, \gamma, d_0)$

trajectory: $\tau = (s_1, a_1, r_1, s_2, a_2, r_2, \cdots)$

$s_1 \sim d_0$
$a_t \sim \pi_\theta(\cdot|s_t)$ (policy)
$r_t \sim R(\cdot|s_t, a_t)$
$s_{t+1} \sim P(\cdot|s_t, a_t)$

Policy optimization: find π to maximize

$$J(\pi) = \mathbb{E}_\pi\left[\sum_{t=1}^{\infty} \gamma^{t-1} r_t\right]$$

value
$$V^\pi(s) = \mathbb{E}_\pi\left[\sum_{t=1}^{\infty} \gamma^{t-1} r_t \,\middle|\, s_1 = s\right]$$
$$Q^\pi(s,a) = \mathbb{E}_\pi\left[\sum_{t=1}^{\infty} \gamma^{t-1} r_t \,\middle|\, s_1 = s, a_1 = a\right]$$

parameterized policy $\{\pi_\theta : \theta \in \Theta\}$

GD: $\theta \leftarrow \theta + \alpha \cdot \nabla_\theta J(\pi_\theta)$

How to compute policy gradient via samples?

wlog. $\tau = (s_1, a_1, r_1, \cdots, s_H, a_H, r_H)$

Policy Gradient:

$$\nabla_\theta J(\pi_\theta) = \nabla_\theta \mathbb{E}_\pi\left[\sum_{t=1}^{H} \gamma^{t-1} r_t\right]$$

$$= \nabla_\theta \mathbb{E}_{\tau\sim\pi}\left[R(\tau)\right] \qquad R(\tau) = \sum_{t=1}^{H} \gamma^{t-1} r_t$$

$$= \nabla_\theta \sum_\tau P^\pi(\tau)\cdot R(\tau)$$

$$= \sum_\tau \left(\nabla_\theta P^\pi(\tau)\right) R(\tau)$$

★ log-trick: $\frac{d}{dx}\log f(x) = \frac{f'(x)}{f(x)}$

$$= \sum_\tau P^\pi(\tau)\cdot \nabla_\theta \log P^\pi(\tau)\cdot R(\tau)$$

$$= \mathbb{E}_{\tau\sim\pi}\left[\nabla_\theta \log P^\pi(\tau)\cdot R(\tau)\right]$$

$$P^\pi(\tau) = d_0(s_1)\cdot \pi_\theta(a_1|s_1)\cdot R(r_1|s_1,a_1)\cdot P(s_2|s_1,a_1)\cdot \pi_\theta(a_2|s_2)\cdots$$

$$\log P^\pi(\tau) = \log d_0(s_1) + \sum_{t=1}^{H} \log \pi_\theta(a_t|s_t) + \sum_{t=1}^{H} \log R(r_t|s_t,a_t) + \sum_{t=1}^{H-1} \log P(s_{t+1}|s_t,a_t)$$

$$\nabla_\theta \log P^\pi(\tau) = \sum_{t=1}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t)$$

"REINFORCE": $$\nabla_\theta J(\pi_\theta) = \mathbb{E}_\pi\left[\left(\sum_{t=1}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t)\right)\left(\sum_{t=1}^{H} \gamma^{t-1} r_t\right)\right]$$

- $\tau \sim \pi_\theta$: $\left(\sum_{t=1}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t)\right)\cdot\left(\sum_{t=1}^{H} \gamma^{t-1} r_t\right)$ Monte-Carlo
- $\tau_1, \tau_2, \cdots, \tau_N \sim \pi_\theta$: $\frac{1}{N}\sum_{i=1}^{N}\left(- - \sim \right)$

↓

on-policy algorithm

Variance reduction

a_t : $r_t, r_{t+1}, \cdots, r_H$

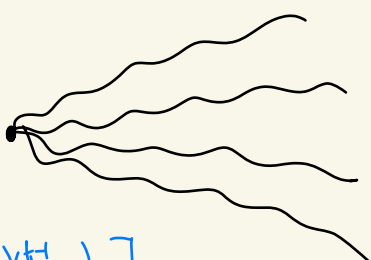
recall: $\theta \leftarrow \theta + \alpha \cdot \nabla_\theta J(\pi_\theta)$

$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_\pi\left[\left(\sum_{t=1}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t)\right)\cdot\left(\sum_{t=1}^{H} \gamma^{t-1} r_t\right)\right]$$

policy score

↓

unbiased ✓ high variance.



$$\nabla_\theta J(\pi_\theta) = \mathbb{E}_\pi\left[\sum_{t=1}^{H} \nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{t=1}^{H} \gamma^{t-1} r_t\right)\right]$$

$$= \sum_{t=1}^{H} \mathbb{E}_\pi\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(\sum_{t'=1}^{t-1} \gamma^{t'-1} r_{t'} + \sum_{t'=t}^{H} \gamma^{t'-1} r_{t'}\right)\right].$$

Claim: $\forall t: \mathbb{E}_\pi\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot \sum_{t'=1}^{t-1} \gamma^{t'-1} r_{t'}\right] = 0$

$$= \mathbb{E}_{\pi}\left[\mathbb{E}_{a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot \sum_{t'=1}^{t-1} \gamma^{t'-1} r_{t'} \,\middle|\, s_1, a_1, r_1, \cdots, s_{t-1}, a_{t-1}, r_{t-1}, s_t\right]\right]$$

law of total expectation.

$$= \mathbb{E}_\pi\left[\sum_{t'=1}^{t-1}\gamma^{t'-1} r_t \cdot \mathbb{E}_{a_t\sim\pi_\theta(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\right]\right]$$

$$= 0$$

$$\Rightarrow \text{causality !}$$

$$= \sum_{a_t} \pi_\theta(a_t|s_t)\cdot \nabla_\theta \log \pi_\theta(a_t|s_t)$$

$$= \sum_{a_t} \pi_\theta(a_t|s_t)\cdot \frac{\nabla_\theta \pi_\theta(a_t|s_t)}{\pi_\theta(a_t|s_t)}$$

$$= \nabla_\theta \sum_{a_t} \pi_\theta(a_t|s_t) = \nabla_\theta 1 = 0,$$

$$\Rightarrow \nabla_\theta J(\pi_\theta)$$

$$= \sum_{t=1}^{H} \mathbb{E}_\pi\left[\nabla_\theta \log \pi(a_t|s_t)\cdot \sum_{t'=t}^{H}\gamma^{t'-1} r_t\right]$$

$$= \sum_{t=1}^{H} \mathbb{E}_{s_t,a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot \gamma^{t-1}\cdot \mathbb{E}_\pi\left[\sum_{t'=t}^{H}\gamma^{t'-t} r_t \,\middle|\, s_t, a_t\right]\right]$$

$$= \sum_{t=1}^{H} \gamma^{t-1}\cdot \mathbb{E}_{s_t,a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot Q^\pi(s_t,a_t)\right]$$

"Actor-Critic"

① Claim: $\forall t$: $\mathbb{E}_{a_t\sim\pi_\theta(\cdot|s_t)}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot b(s_t)\right] = 0$

$b(s_t)$: baseline — independent to a_t.

$$\nabla_\theta J(\pi_\theta) = \sum_{t=1}^{H} \gamma^{t-1}\cdot \mathbb{E}_{s_t,a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\left(Q^\pi(s_t,a_t) - b(s_t)\right)\right]$$

$b(s_t) = V^\pi(s_t)$, $Q^\pi(s_t,a_t) - V^\pi(s_t) = A^\pi(s_t,a_t)$: advantage.

$$\nabla_\theta J(\pi_\theta) = \sum_{t=1}^{H} \gamma^{t-1}\cdot \mathbb{E}_{s_t,a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t)\cdot A^\pi(s_t,a_t)\right]$$

② $\nabla_\theta J(\pi_\theta) \approx \sum_{t=1}^{H} \mathbb{E}_{s_t, a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A^\pi(s_t, a_t)\right]$

↑ In practice.

Advantage estimation

recall: $\nabla_\theta J(\pi_\theta) = \sum_{t=1}^{H} \gamma^{t-1} \mathbb{E}_{s_t, a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A^\pi(s_t, a_t)\right]$

advantage

$G_t = \sum_{t'=t}^{\infty} \gamma^{t'-t} r_t$: reward-to-go.

$A^\pi(s,a) = Q^\pi(s,a) - V^\pi(s)$

$= Q^\pi(s,a) - Q^\pi(s,\pi)$

$\Rightarrow \begin{cases} Q^\pi(s,a) = \mathbb{E}_\pi[G_t | s_t = s, a_t = a] \\ V^\pi(s) = \mathbb{E}_\pi[G_t | s_t = s] \end{cases}$

Assume we have some learned representation $\hat{V}_\phi \approx V^\pi$

critic

1) Monte-Carlo: $\hat{A}_t^{MC} = G_t - \hat{V}_\phi(s_t)$

2) Temporal Difference (TD): $Q^\pi(s_t, a_t) = \mathbb{E}[r_t + \gamma \cdot V^\pi(s_{t+1})]$

$A^\pi(s_t, a_t) = Q^\pi(s_t, a_t) - V^\pi(s_t)$

$= \mathbb{E}[r_t + \gamma \cdot V^\pi(s_{t+1}) - V^\pi(s_t)]$

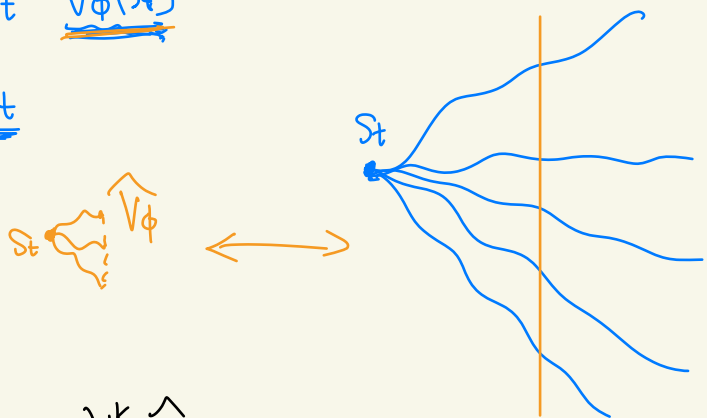
def (TD error) $\delta_t = r_t + \gamma \cdot \hat{V}_\phi(s_{t+1}) - \hat{V}_\phi(s_t)$

if $\hat{V}_\phi = V^\pi$: $\mathbb{E}[\delta_t | s_t, a_t] = A^\pi(s_t, a_t)$

$\Rightarrow$ one-step advantage estimator: $\hat{A}_t^{(1)} = \delta_t$.

MC: $\hat{A}_t^{MC} = G_t - \hat{V}_\phi(s_t)$

TD: $\hat{A}_t^{(1)} = \delta_t$



k-step return:

$$\hat{G}_t^{(k)} = \sum_{t'=t}^{t+k-1} \gamma^{t'-t} r_{t'} + \gamma^k \cdot \hat{V}_\phi(s_{t+k})$$

$\Rightarrow$ k-step advantage estimator: $\hat{A}_t^{(k)} = \hat{G}_t^{(k)} - \hat{V}_\phi(s_t)$.

e.g. $\hat{A}_t^{(1)} = r_t + \gamma \cdot \hat{V}_\phi(s_{t+1}) - \hat{V}_\phi(s_t) = \delta_t$.

$$\hat{A}_t^{(2)} = r_t + \gamma \cdot r_{t+1} + \gamma^2 \cdot \hat{V}_\phi(s_{t+2}) - \hat{V}_\phi(s_t),$$

$$= \left(r_t + \gamma \cdot \hat{V}_\phi(s_{t+1}) - \hat{V}_\phi(s_t)\right) + \gamma \cdot \left(r_{t+1} + \gamma \cdot \hat{V}_\phi(s_{t+2}) - \hat{V}_\phi(s_{t+1})\right)$$

$$= \delta_t + \gamma \cdot \delta_{t+1}.$$

$$\hat{A}_t^{(3)} = \delta_t + \gamma\cdot\delta_{t+1} + \gamma^2\cdot\delta_{t+2}$$

$$\vdots$$

$$\hat{A}_t^{(k)} = \sum_{t'=t}^{t+k-1} \gamma^{t'-t}\cdot\delta_{t'}$$

Generalized Advantage Estimator (GAE)

λ : eligibility trace.

$$\hat{A}_t^{GAE} = (1-\lambda)\cdot\sum_{k=1}^{\infty} \lambda^{k-1}\cdot \hat{A}_t^{(k)}$$

$$= (1-\lambda)\cdot\left(\delta_t + \lambda\cdot(\delta_t+\gamma\cdot\delta_{t+1}) + \lambda^2(\delta_t+\gamma\cdot\delta_{t+1}+\gamma^2\cdot\delta_{t+2}) + \cdots\right)$$

$$= \delta_t\cdot(1-\lambda)\cdot(1+\lambda+\lambda^2+\cdots) \quad \to 1$$

$$+ \gamma\cdot\delta_{t+1}\cdot(1-\lambda)\cdot(\lambda+\lambda^2+\cdots) \quad \to \lambda$$

$$+ \gamma^2\cdot\delta_{t+2}(1-\lambda)(\lambda^2+\lambda^3+\cdots) \quad \to \lambda^2$$

$$+ \cdots$$

$$= \delta_t + \gamma\lambda\cdot\delta_{t+1} + (\gamma\lambda)^2\delta_{t+2} + \cdots$$

$$\Rightarrow \hat{A}_t^{GAE} = \sum_{t'=t}^{\infty} (\gamma\lambda)^{t'-t}\cdot\delta_{t'}$$

$$\hat{A}_t^{GAE} = \delta_t + \gamma\cdot\lambda\cdot\hat{A}_{t+1}^{GAE} \quad : \text{from } H \to 1$$

$\lambda = 0$: $\hat{A}_t^{GAE} = \delta_t = \hat{A}_t^{(1)}$

$\lambda = 1$: $\hat{A}_t^{GAE} = \sum_{t'=t}^{\infty} \gamma^{t'-t} \left(r_{t'} + \gamma \hat{V}_\phi(s_{t+1}) - \hat{V}_\phi(s_t) \right)$

$= \sum_{t'=t}^{\infty} \gamma^{t'-t} r_{t'} - \hat{V}_\phi(s_t)$

Where do we learn $\hat{V}_\phi$?

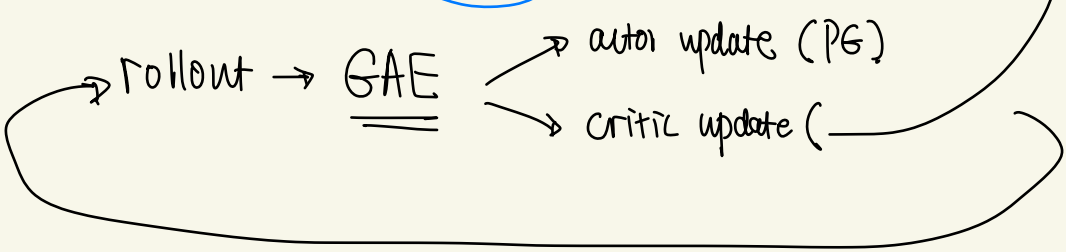
$\hat{V}_\phi(s_t) \approx V^{\pi_\theta}(s_t) = \mathbb{E}_\pi[G_t | s_t]$

- MC : $\min_\phi \mathbb{E}_{s_t}\left[\left(\hat{V}_\phi(s_t) - G_t \right)^2 \right]$

- Bootstrapping : (TD) : $\min_\phi \mathbb{E}_{s_t}\left[\left(\hat{V}_\phi(s_t) - r_t - \gamma \hat{V}_{\phi_{old}}(s_{t+1}) \right)^2 \right]$

- (GAE) : $\min_\phi \mathbb{E}_{s_t}\left[\left(\hat{V}_\phi(s_t) - \hat{A}_t^{GAE} - \hat{V}_{\phi_{old}}(s_{t+1}) \right)^2 \right]$ (*)

off-policy ; asynchronous update !



Recall: Actor-Critic

At round k:

1) rollout trajectories $\{\tau^{(i)}\}_{i=1}^{N} \sim \pi_{\theta_k}$
2) fit $\hat{V}_\phi \approx V^{\pi_{\theta_k}}$
3) evaluate $\hat{A}_t^{GAE} \approx A^{\pi_{\theta_k}}$
4) compute $\nabla_\theta J(\pi_{\theta_k}) \approx \frac{1}{N}\sum_{i=1}^{N}\sum_{t=1}^{H} \nabla_\theta \log \pi_{\theta_k}(a_t^{(i)}|s_t^{(i)}) \cdot \hat{A}_t^{GAE}(s_t^{(i)}, a_t^{(i)})$
5) $\theta_{k+1} \leftarrow \theta_k + \alpha \cdot \nabla_\theta \hat{J}(\pi_{\theta_k})$

(multiple times)

PG: on-policy → off-policy (sample efficiency ↑)

$$\nabla_\theta J(\pi_\theta) = \sum_{t=1}^{H} \gamma^t \cdot \mathbb{E}_{s_t, a_t}\left[\nabla_\theta \log \pi_\theta(a_t|s_t) \cdot A^{\pi_\theta}(s_t, a_t)\right]$$

$s_t \sim d^{\pi_\theta}, a_t \sim \pi_\theta(\cdot|s_t)$ mismatch ⇒ correct

① $s_t \sim d^{\pi_k}, a_t \sim \pi_k(\cdot|s_t)$ ② $A^{\pi_k}(s_t, a_t)$

def (occupancy distribution) $d^\pi \in \Delta(S)$

$$d^\pi(s) = (1-\gamma)\cdot\sum_{t=1}^{\infty}\gamma^{t-1}\cdot d_t^\pi(s)$$

$$= (1-\gamma)\cdot\sum_{t=1}^{\infty}\gamma^{t-1}\cdot P^\pi(s_t = s)$$

PD Lemma (Performance Difference Lemma)

$$J(\pi_\theta) - J(\pi_k) = \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_\theta}, a\sim\pi_\theta(\cdot|s)}\left[A^{\pi_k}(s,a)\right]$$

$\pi_\theta = \pi_k$: $\mathbb{E}_{a\sim\pi_k(\cdot|s)}[A^{\pi_k}(s,a)]$

$= \mathbb{E}_{a\sim\pi_k(\cdot|s)}[Q^{\pi_k}(s,a) - V^{\pi_k}(s)] = 0.$

Proof of PD Lemma :

$$A^{\pi_k}(s_t,a_t) = \mathbb{E}\left[r_t + \gamma\cdot V^{\pi_k}(s_{t+1}) - V^{\pi_k}(s_t) \mid s_t, a_t\right]$$

$$\text{RHS} = \mathbb{E}_{\pi_\theta}\left[\sum_{t=1}^{\infty}\gamma^{t-1}\cdot A^{\pi_k}(s_t,a_t)\right] = \mathbb{E}_{\pi_\theta}\left[\sum_{t=1}^{\infty}\gamma^{t+1}r_t - V^{\pi_k}(s_1)\right]$$

$$= J(\pi_\theta) - J(\pi_k). \quad \square$$

first order

$$\nabla_\theta J(\pi_\theta) = \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_\theta},\, a\sim\pi_\theta(\cdot|s)}\left[\nabla_\theta \log\pi_\theta(a|s)\cdot A^{\pi_\theta}(s,a)\right]$$

π_θ : $s\sim d^{\pi_\theta}$ · $a\sim\pi_\theta(\cdot|s)$ → p

π_k : data: $s\sim d^{\pi_k}$ · $a\sim\pi_k(\cdot|s)$ → q

Heuristic : assume $d^{\pi_\theta}(s) \approx d^{\pi_k}(s)$

(when π_θ is close to π_k)

Importance sampling

target dist: p.

source dist: q.

$$\mathbb{E}_p[f] = \sum_x p(x) f(x)$$

$$= \sum_x q(x)\left[\frac{p(x)}{q(x)}\right] f(x)$$

$$= \mathbb{E}_q\left[\frac{p}{q}\cdot f\right].$$

↑ estimator

$$\frac{d^{\pi_\theta}(s)\cdot\pi_\theta(a|s)}{d^{\pi_k}(s)\cdot\pi_k(a|s)} \approx \frac{\pi_\theta(a|s)}{\pi_k(a|s)} := \rho_k(\theta; s, a)$$

IS ratio

Surrogate objective:

$$L_k(\theta) = \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\rho_k(\theta;s,a)\cdot A^{\pi_k}(s,a)\right]$$

when $\theta \approx \theta_k$: $\nabla_\theta L_k(\theta)\big|_{\theta_k} \approx \nabla_\theta J(\pi_k)$

$$\nabla_\theta \rho_k(\theta; s,a) = \nabla_\theta \frac{\pi_\theta(a|s)}{\pi_k(a|s)}$$

$$= \frac{\pi_\theta(a|s)}{\pi_k(a|s)} \cdot \nabla_\theta \log \pi_\theta(a|s)$$

$$= \rho_k(\theta; s,a) \cdot \nabla_\theta \log \pi_\theta(a|s)$$

When $\theta = \theta_k$: $\rho_k(\theta; s,a) = 1$. $L_k(\theta_k) = 0$.

$$\Rightarrow \nabla_\theta L_k(\theta)\big|_{\theta_k} = \frac{1}{1-\gamma} \cdot \mathbb{E}_{\substack{s\sim d^{\pi_k} \\ a\sim\pi_k(\cdot|s)}} \left[A^{\pi_k}(s,a) \right]$$

$$= \nabla_\theta J(\pi_k)$$

$\Rightarrow L_k$ is first-order approximation of policy improvement $J(\theta) - J(\theta_k)$ around $\theta = \theta_k$.

A natural strategy: $\max_\theta L_k(\theta)$

→ multiple updates: $d^{\pi_\theta} \approx d^{\pi_k}$

↓ constrain updates!



TRPO

$$\max_\theta L_k(\theta)$$

$$\text{s.t. } \mathbb{E}_{s\sim d^{\pi_k}} \left[KL\left(\pi_k(\cdot|s) \,\|\, \pi_\theta(\cdot|s)\right) \right] \le \delta.$$

trust region

$P, Q \in \Delta(X)$: $KL(P\|Q) = \sum_x P(x) \cdot \log \frac{P(x)}{Q(x)}$.

$KL(P\|Q) \ge 0$. and $KL(P\|Q) = 0$ iff $P = Q$.

⇓

Proximal Policy Optimization. (PPO)

PPO-KL-penalty :

$$L_k^{KL}(\theta) = \mathbb{E}_{\substack{s\sim d^{\pi_k} \\ a\sim\pi_k(\cdot|s)}}\left[\rho_k(\theta;s,a)\cdot A^{\pi_k}(s,a) - \beta_k \cdot KL\left(\pi_k(\cdot|s)\,\|\,\pi_\theta(\cdot|s)\right)\right]$$

β_k: adaptive.

Recall: "constrained" policy optimization.

$$J(\pi_\theta) - J(\pi_k) = \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_\theta},\, a\sim\pi_\theta(\cdot|s)}\left[A^{\pi_k}(s,a)\right]$$

$$\approx \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\frac{\pi_\theta(a|s)}{\pi_k(a|s)}\cdot A^{\pi_k}(s,a)\right] \quad \text{(IS)}$$

If π_θ is closed to π_k: $d^{\pi_\theta}(s) \approx d^{\pi_k}(s)$

PPO-clipping.

$$L_k(\theta) = \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\rho_k(\theta;s,a)\cdot A^{\pi_k}(s,a)\right]$$

$\rho_k(\theta;s,a)$: IS ratio.

def (clipping)

$$\text{clip}(\rho, 1-\varepsilon, 1+\varepsilon) = \begin{cases} 1-\varepsilon, & \rho < 1-\varepsilon \\ \rho, & 1-\varepsilon \le \rho \le 1+\varepsilon \\ 1+\varepsilon, & \rho > 1+\varepsilon \end{cases}$$

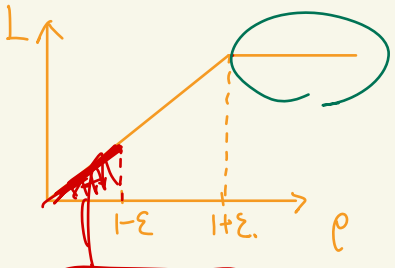
PPO-CLIP:

pessimism

$$L_k^{CLIP}(\theta) = \mathbb{E}_{d^{\pi_k}}\left[\min\left\{\rho_k \cdot A^{\pi_k},\ clip(\rho_k, 1-\varepsilon, 1+\varepsilon)\cdot A^{\pi_k}\right\}\right]$$

① $A^{\pi_k}(s,a) > 0$: good action.

$$L_k^{CLIP}(\theta) = \mathbb{E}_{d^{\pi_k}}\left[\min\{\rho_k, 1+\varepsilon\}\cdot A^{\pi_k}\right]$$



Large improvement → clip

$\frac{dL}{d\rho} = A > 0$,

$\rho \uparrow \Rightarrow L \uparrow$

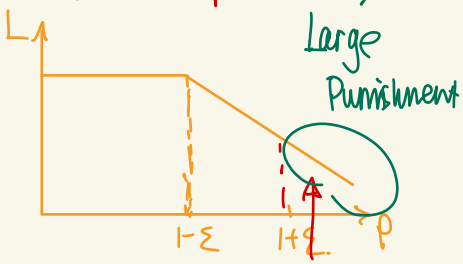
$$\nabla_\theta L = A \cdot \nabla_\theta \rho = \rho A \cdot \nabla_\theta \log \pi$$

$\frac{dL}{d\rho} = A < 0$.

$\rho \downarrow \quad L \uparrow$

keep

Large Punishment



② $A^{\pi_k}(s,a) < 0$: bad action.

$$L_k^{CLIP}(\theta) = \mathbb{E}_{d^{\pi_k}}\left[\max\{\rho_k, 1-\varepsilon\}\cdot A^{\pi_k}\right]$$

Favorable change: clipping (over-reward)

Unfavorable change: not clipping

A. + entropy regularization (optional)

$+ c_H \cdot \mathbb{E}_{s\sim d^{\pi_k}}[\mathcal{H}(\pi_\theta(\cdot|s))]$

$p \in \Delta(\mathcal{X})$

$\mathcal{H}(p) = -\sum_x p(x)\log p(x)$

↑ entropy.

---

1. Rollout.
2. GAE.
3. Critic → multi-step
4. actor → multi-step
5. update parameters.

$$\nabla_\theta J(\theta) = \frac{1}{1-\gamma}\cdot \mathbb{E}_{d^{\pi_\theta}}[\nabla \log \pi \cdot A^\pi]$$

$$= \mathbb{E}_\pi\left[\sum_{t=1}^{H} \gamma^{t-1} \cdot \nabla \log \pi \cdot A^\pi\right]$$

e.g.

$$\sum_{t=1}^{H} \min\{\rho_k A^{\pi_k}, \text{clip}(\rho_k, 1-\varepsilon, 1+\varepsilon) A^{\pi_k}\}$$

PPO-KL-Penalty: $L_k^{KL}(\theta) = \mathbb{E}_{s\sim d^{\pi_k}, a\sim\pi_k(\cdot|s)}\left[\frac{\pi_\theta(a|s)}{\pi_k(a|s)} A^{\pi_k}(s,a) - \beta_k KL(\pi_k \| \pi_\theta)\right]$

PPO-CLIP: $L_k^{CLIP}(\theta) = \mathbb{E}_{d^{\pi_k}}[\min\{\rho_k A^{\pi_k}, \text{clip}(\rho_k, 1-\varepsilon, 1+\varepsilon) A^{\pi_k}\}]$

Intuition: if π_θ is close to π_k. $d^{\pi_\theta}(s) \approx d^{\pi_k}(s)$

Example. π_k deterministic.

"close" if $\pi_\theta(a_t \neq \pi_k(s_t) | s_t) \leq \varepsilon$. $\forall t$.

$d_t^{\pi}(s) = \mathbb{P}^{\pi}[S_t = s]$.

$$d_t^{\pi_\theta}(S_t) = (1-\varepsilon)^{t-1} d_t^{\pi_k}(S_t) + \left(1-(1-\varepsilon)^{t-1}\right) \cdot P_{\text{mistake}}(S_t)$$

prob

some other dist.

$$\Rightarrow d_t^{\pi_\theta}(S_t) - d_t^{\pi_k}(S_t) = \left(1-(1-\varepsilon)^{t-1}\right)\left(P_{\text{mistake}}(S_t) - d_t^{\pi_k}(S_t)\right)$$

$$\frac{1}{2}\sum_{S_t}\left|d_t^{\pi_\theta}(S_t) - d_t^{\pi_k}(S_t)\right| = \frac{1}{2}\left(1-(1-\varepsilon)^{t-1}\right)\cdot\sum_{S_t}\left|P_{\text{mistake}}(S_t) - d_t^{\pi_k}(S_t)\right|$$

$$\leq 1-(1-\varepsilon)^{t-1}$$

$$\leq \varepsilon(t-1)$$

$(1-\varepsilon)^t \geq 1-\varepsilon t. \quad \forall \varepsilon \in [0,1]$

$$d^{\pi}(s) = (1-\gamma)\cdot\sum_{t=1}^{\infty}\gamma^{t-1} d_t^{\pi}(s)$$

$$\frac{1}{2}\sum_{s}\left|d^{\pi_\theta}(s) - d^{\pi_k}(s)\right| \leq (1-\gamma)\cdot\sum_{t=1}^{\infty}\gamma^{t-1}\varepsilon\cdot(t-1)$$

$$\leq \frac{2\varepsilon}{1-\gamma}.$$

General:

"closeness": $\forall t, S_t$: $\frac{1}{2}\sum_{a_t}\left|\pi_\theta(a_t|S_t) - \pi_k(a_t|S_t)\right|$

$$= \frac{1}{2}\left\|\pi_\theta(\cdot|S_t) - \pi_k(\cdot|S_t)\right\|_1$$

$$= TV\left(\pi_\theta(\cdot|S_t), \pi_k(\cdot|S_t)\right) \leq \varepsilon.$$

If P_X, P_Y: $TV(P_X, P_Y) \le \varepsilon$.

$\exists$ joint distribution $p(x,y)$ s.t. $\begin{cases} p(x) = P_X(x) \\ p(y) = P_Y(y) \end{cases}$

and $p(x=y) \ge 1-\varepsilon$.

coupling

$\Rightarrow$ $P_k(x)$ "agrees" with $P_Y(y)$ w.p. $\ge 1-\varepsilon$.

$\Rightarrow$ $\pi_\theta(\cdot|s_t)$ takes a different action than $\pi_k(\cdot|s_t)$ w.p. $\ge 1-\varepsilon$.

$\vdots$

$$\Rightarrow TV(d^{\pi_\theta}, d^{\pi_k}) = \frac{1}{2}\sum_s |d^{\pi_\theta}(s) - d^{\pi_k}(s)| \le \frac{\gamma\varepsilon}{1-\gamma}.$$

$\forall f$ bounded in $[0, V_{max}]$

$$\mathbb{E}_{s\sim d^{\pi_\theta}}[f(s)] = \sum_s d^{\pi_\theta}(s)\cdot f(s)$$

$$\ge \sum_s d^{\pi_k}(s)\cdot f(s) - |d^{\pi_\theta}(s) - d^{\pi_k}(s)| \cdot \max_s |f(s)|$$

$$\ge \mathbb{E}_{s\sim d^{\pi_k}}[f(s)] - \frac{\gamma\varepsilon}{1-\gamma}\cdot V_{max}$$

PD Lemma: $J(\pi_\theta) - J(\pi_k) = \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_\theta}, a\sim\pi_\theta(\cdot|s)}[A^{\pi_k}(s,a)]$

$= \frac{1}{1-\gamma}\cdot \mathbb{E}_{s\sim d^{\pi_\theta}}[A^{\pi_k}(s,\pi_\theta)]$

policy improvement

$$\geq \frac{1}{1-\gamma} \cdot \mathbb{E}_{s\sim d^{\pi_k}}\left[A^{\pi_k}(s,\pi_\theta)\right] - \frac{\gamma\varepsilon}{(1-\gamma)^2} V_{max}$$

$$= \frac{1}{1-\gamma}\left(L_k(\theta) - \frac{\gamma\varepsilon}{1-\gamma} V_{max}\right)$$

as long as. $TV\left(\pi_\theta(\cdot|s), \pi_k(\cdot|s)\right) \leq \varepsilon$.

$\Rightarrow KL\left(\pi_k(\cdot|s) \,\|\, \pi_\theta(\cdot|s)\right) \leq \varepsilon'$.

for small enough ε: gradient update

→ monotonic improvement. guarantee.

Pinsker's ineq: $p, q \in \Delta(\mathcal{X})$. $TV(p,q) \leq \sqrt{\frac{1}{2} KL(p\|q)}$

TRPO, NPG.

recall: PPO-KL-penalty $L_k^{KL}(\theta) = L_k(\theta) - \beta_k \cdot D_k(\theta)$

- $L_k(\theta) = \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\frac{\pi_\theta(a|s)}{\pi_k(a|s)} \cdot A^{\pi_k}(s,a)\right]$
- $D_k(\theta) = \mathbb{E}_{s\sim d^{\pi_k}}\left[KL\left(\pi_k(\cdot|s) \,\|\, \pi_\theta(\cdot|s)\right)\right]$

Approximation: $\Delta = \theta - \theta_k$

① $\nabla_\theta L_k(\theta) = \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\frac{\pi_\theta(a|s)}{\pi_k(a|s)} \cdot \nabla_\theta \log \pi_\theta(a|s) \cdot A^{\pi_k}(s,a)\right]$

$g_k = \nabla_\theta L_k(\theta)\big|_{\theta_k} = \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_k(a|s) \cdot A^{\pi_k}(s,a)\right]$

$$= (1-\gamma)\cdot \nabla_\theta J(\pi_\theta)\big|_{\theta_k}.$$

$$L_k(\theta_k) = 0$$

$$L_k(\theta) = L_k(\theta_k) + g_k^T \Delta + O(\|\Delta\|_2^2).$$

$$\approx g_k^T \Delta.$$

② $D_k(\theta_k) = 0.$

$$\nabla_\theta D_k(\theta) = \mathbb{E}_{s\sim d^{\pi_k}}\left[\nabla_\theta \sum_a \pi_k(a|s)\cdot \log \frac{\pi_k(a|s)}{\pi_\theta(a|s)}\right]$$

$$= \mathbb{E}_{s\sim d^{\pi_k}}\left[-\sum_a \pi_k(a|s)\cdot \nabla_\theta \log \pi_\theta(a|s)\right]$$

$$= -\mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s)\right] \quad (*)$$

$$\nabla_\theta D_k(\theta)\big|_{\theta_k} = -\mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_k(a|s)\right] = 0.$$

$$D_k(\theta) = \frac{1}{2}\Delta^T \nabla_\theta^2 D_k(\theta)\big|_{\theta_k}\Delta + O(\|\Delta\|_2^2)$$

$$\nabla_\theta^2 D_k(\theta) = -\mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta^2 \log \pi_\theta(a|s)\right]$$

$$= \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_\theta(a|s)\cdot \nabla_\theta \log \pi_\theta(a|s)^T\right]$$

$$\Downarrow$$

$$\nabla_\theta^2 D_k(\theta)\big|_{\theta_k} := F_k$$

$$= \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_k(a|s)\cdot \nabla_\theta \log \pi_k(a|s)^T\right]$$

: Fisher information matrix

Fisher information identity

$P_\theta \in \Delta(\mathcal{X})$

$$\mathbb{E}_{P_\theta}\left[\nabla^2 \log P_\theta\right] = -\mathbb{E}_{P_\theta}\left[(\nabla \log P_\theta)(\nabla \log P_\theta)^T\right]$$

Proof: $\mathbb{E}_{P_\theta}\left[\nabla \log P_\theta\right] = \sum_x \nabla P_\theta(x) = 0.$

$$\Rightarrow \sum_x \nabla^2 P_\theta(x) = 0.$$

$$\nabla^2 p_\theta(x) = \nabla\left(\nabla p_\theta(x)\right) = \nabla\left(p_\theta(x)\,\nabla \log p_\theta(x)\right)$$

$$= \nabla p_\theta(x)\cdot \nabla \log p_\theta(x)^T + p_\theta(x)\cdot \nabla^2 \log p_\theta(x)$$

$$= p_\theta(x)\cdot \nabla \log p_\theta(x)\cdot \nabla \log p_\theta(x)^T + p_\theta(x)\cdot \nabla^2 \log p_\theta(x)$$

$$\Rightarrow 0 = \sum_x \nabla^2 p_\theta(x) = \sum_x p_\theta(x)\cdot\left(\nabla \log p_\theta(x)\cdot \nabla \log p_\theta(x)^T + \nabla^2 \log p_\theta(x)\right)$$

$$\Rightarrow \begin{cases} L_k(\theta) \approx g_k^T \Delta \\ D_k(\theta) \approx \frac{1}{2}\Delta^T F_k \Delta \end{cases}$$

$$\Rightarrow \max_\Delta \left\{ g_k^T \Delta - \frac{\beta_k}{2}\cdot \Delta^T F_k \Delta \right\}$$

quadratic function

$$g_k - \beta_k F_k \cdot \Delta^* = 0 \Rightarrow \Delta^* = \frac{1}{\beta_k}\cdot F_k^{-1} g_k$$

pre-conditioner.

geometry

≈ KL divergence.

$KL(\pi_k \| \pi_\theta)$

↳ policy space.

Natural Policy Gradient (NPG):

$$\theta_{k+1} = \theta_k + \alpha_k \cdot F_k^{-1} g_k \quad \Leftrightarrow$$

FIM

$$\max_\Delta g_k^T \Delta \quad \text{s.t. } \frac{1}{2}\Delta F_k \Delta \le \delta$$

Vanilla Policy Gradient:

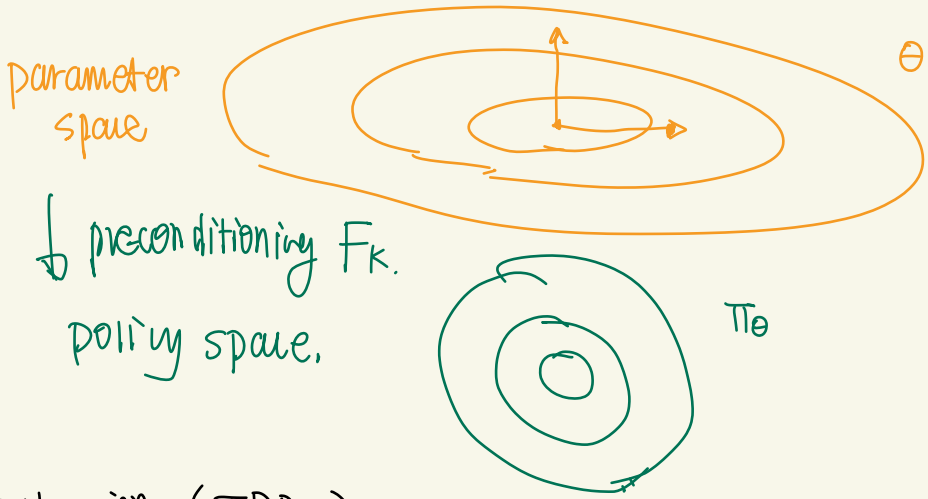
Euclidean norm

$\|\theta_k - \theta\|_2$

↳ parameter space.

$$\theta_{k+1} = \theta_k + \alpha_k \cdot g_k \quad \Leftrightarrow$$

$$\max_\Delta g_k^T \Delta \quad \text{s.t. } \|\Delta\|_2^2 \le \delta.$$



trust region : (TRPO)

$$\max_{\theta} L_k(\theta) \quad \text{s.t.} \quad D_k(\theta) \le \delta$$

$$\Rightarrow \max_{\Delta} g_k^T \Delta \quad \text{s.t.} \quad \frac{1}{2}\Delta^T F_k \Delta \le \delta .$$

Lagrangian: $\mathcal{L}(\Delta, \lambda) = g_k^T \Delta - \lambda \cdot \left(\frac{1}{2}\Delta^T F_k \Delta - \delta\right)$

$$\Rightarrow g_k - \lambda \cdot F_k \cdot \Delta = 0$$

$$\Rightarrow \Delta = \frac{1}{\lambda} \cdot F_k^{-1} g_k .$$

$$\Rightarrow \frac{1}{\lambda} g_k^T F_k^{-1} g_k - \frac{1}{2\lambda} g_k^T F_k^{-1} g_k + \lambda\delta = 0$$

$$\Rightarrow \lambda = \sqrt{\frac{g_k^T F_k^{-1} g_k}{2\delta}}$$

$$\Rightarrow \Delta^* = \sqrt{\frac{2\delta}{g_k^T F_k^{-1} g_k}} \cdot F_k^{-1} g_k .$$

TRPO: $\theta_{k+1} = \theta_k + \alpha^j \sqrt{\frac{2\delta}{g_k^T F_k^{-1} g_k}} F_k^{-1} g_k.$

backtracking line search. $0<\alpha<1$. $j=0$

$$\begin{cases} g_k = \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_k(a|s) \cdot A^{\pi_k}(s,a)\right] \\ F_k = \mathbb{E}_{s\sim d^{\pi_k},\, a\sim\pi_k(\cdot|s)}\left[\nabla_\theta \log \pi_k(a|s) \cdot \nabla_\theta \log \pi_k(a|s)^T\right] \end{cases}$$

Natural gradient: $F_k^{-1} g_k = \mathbb{E}[xx^T]^{-1} \cdot \mathbb{E}[xy]$

$x = \nabla_\theta \log \pi_k(a|s)$

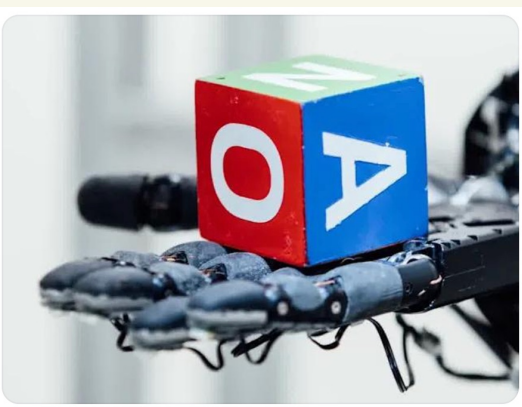
$y = A^{\pi_k}(s,a)$

$w \in \operatorname{argmin} \mathbb{E}\left[(w^T x - y)^2\right]$

actor ↔ critic

$w^T \nabla_\theta \log \pi_k(a|s) \sim A^{\pi_k}(s,a)$

compatible function approximation.



OpenAI Dactyl (2018)



OpenAI Five (2019)

LLM-RL

← RLHF
RLVR
Agentic RL

LLM training: ① pre-training
② supervised fine-tuning (SFT)
③ RL post-training

① pre-training:

$D = \{x^{(i)}\}_{i=1}^{n}$. each sequence includes T_i tokens.

Maximum likelihood: $\max_\theta \mathbb{E}_{x\sim D}\left[\sum_{t=1}^{T_i} \log P_\theta(x_t | x_{<t})\right]$

autoregressive:

$P_\theta(x_{1:T_i}) = P_\theta(x_1)\cdot P_\theta(x_2|x_1)$
$\cdot P_\theta(x_3|x_1,x_2)\cdots$
$\cdot P_\theta(x_{T_i}|x_1,x_2,\cdots,x_{T_i-1})$.

② SFT (Imitation Learning)

$D_{SFT} = \{(x_i, y_i^*)\}_{i=1}^{n}$.

y_i^*: high-quality demonstration

$\max_\theta \mathbb{E}_{(x,y)\sim D_{SFT}}[\log \pi_\theta(y^*|x)]$.

← behavior cloning

③ RL : $\max_{\theta} J(\theta) = \mathbb{E}_{x \sim D,\ y \sim \pi_\theta(\cdot|x)} [R(x, y)]$

reward maximization.

x: prompt.

y: LLM-generated response.

π_θ: LLM (policy)

R: reward.

Viewpoint 1: x: state (context)

y: action

↓

1-step MDP [contextual bandits]

$$\pi_\theta(y|x) = \prod_{t=1}^{T} p_\theta(y_t | x, y_{<t})$$

$$\log \pi_\theta(y|x) = \sum_{t=1}^{T} \log p_\theta(y_t | x, y_{<t})$$

Viewpoint 2 : token-level MDP (finite-horizon)

at time t: { State $s_t := (x, y_1, y_2, \cdots, y_{t-1})$

action $a_t = y_t$.

policy p_θ.

transition: deterministic augmentation,

$s_{t+1} = (s_t, a_t)$

reward $r_t = \begin{cases} 0, & \text{if } t<T \\ R, & \text{if } t=T \end{cases}$ (sparse)

↳ credit assignment

initial state distribution $x \sim \mathcal{D}$.

Regime
- transition
- action space.
- ---
- reference model ☆

LLM-RL ⟷ Deep RL.

Policy Gradient (e.g. REINFORCE)

$$J(\theta) = \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi_\theta(\cdot|x)}[R(x,y)]$$

$$\nabla_\theta J(\theta) = \mathbb{E}_{x,y}[R(x,y)\cdot\nabla_\theta \log\pi_\theta(y|x)]$$

$$= \mathbb{E}_{x,y}\Big[R(x,y)\cdot\sum_{t=1}^{T}\nabla_\theta\log\pi_\theta(y_t|s_t)\Big]$$

Reference model π_{ref} (e.g. SFT checkpoint)

π_θ : current policy

π_{ref} : reference policy

π_{old} : behavior policy

$\neq$

"reward hacking"

regularized objective:

$$\max_\theta J_\beta(\theta) = \mathbb{E}_{x\sim D}\left[\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}[R(x,y)] - \beta \cdot KL(\pi_\theta(\cdot|x)\,\|\,\pi_{ref}(\cdot|x))\right]$$

(reverse KL)

* forward KL vs reverse KL

p: model q: target

* PPO: π_{old} vs π_θ.
  LLM-RL: π_θ vs π_{ref}.

reverse KL: $KL(p\|q) = \sum_x p(x)\log\frac{p(x)}{q(x)}$

$= \mathbb{E}_{x\sim p}\left[\log\frac{p(x)}{q(x)}\right]$.

→ mode-seeking.

$p\uparrow, q\downarrow \Rightarrow \frac{p}{q}\to\infty$

forward KL: $KL(q\|p) = \mathbb{E}_{x\sim q}\left[\log\frac{q(x)}{p(x)}\right]$.

→ mode covering

$q\uparrow, p\downarrow \Rightarrow \frac{q}{p}\to\infty$



$$\log\frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)} = \sum_{t=1}^{T}\log\frac{\pi_\theta(y_t|s_t)}{\pi_{ref}(y_t|s_t)}$$

→ regularized reward: $\tilde{r}_t = r_t - \beta\left(\log\pi_\theta(y_t|s_t) - \log\pi_{ref}(y_t|s_t)\right)$

Sample prompts → Generate rollouts → Evaluate rewards → asynchronous

→ Estimate advantages → Policy update → Sync & Update.

Reinforcement Learning from Human Feedback (RLHF)

$x \rightarrow y_A \quad R(x, y_A) = 8.7$

$\rightarrow y_B \quad R(x, y_B) = 5.2$

$A \succ B$

"numerical reward" "preference"

Human preference dataset: (x, y^+, y^-)

- 1) reward modeling : $(x, y^+, y^-) \longrightarrow r_\phi(x, y)$
- 2) policy optimization : $\max_\theta \mathbb{E}_{\pi_\theta}[r_\phi(x, y)] - \beta \cdot KL$.

"InstructGPT"

Reward Modeling

x : $y^{(1)}, \cdots, y^{(k)} \sim \pi_{ref}(\cdot|x)$

e.g. $y^{(1)} \succ y^{(3)} \succ y^{(2)} \Rightarrow (x, y^{(1)}, y^{(3)}), (x, y^{(1)}, y^{(2)}), (x, y^{(3)}, y^{(2)})$

$\Rightarrow D_{pref} = \{(x_i, y_i^+, y_i^-)\}_{i=1}^n$

Goal: $r_\phi : X \times Y \rightarrow \mathbb{R}$

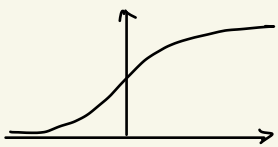
Bradley-Terry (BT) model:

$$P_\phi(y^+ \succ y^- \mid x) = \frac{\exp(r_\phi(x,y^+))}{\exp(r_\phi(x,y^+)) + \exp(r_\phi(x,y^-))}$$

$$= \sigma\left(r_\phi(x,y^+) - r_\phi(x,y^-)\right)$$

reward difference

$$\sigma(z) = \frac{1}{1+e^{-z}}.$$



⇓

Maximum Likelihood estimation:

$$\min_\phi \mathcal{L}_{RM}(\phi) = -\frac{1}{n}\sum_{i=1}^{n} \log \sigma\left(r_\phi(x_i,y_i^+) - r_\phi(x_i,y_i^-)\right)$$

$$\tilde{r}(x,y) = r(x,y) + c(x)$$

$$\mathbb{E}_{y\sim\pi(\cdot|x)}\left[r(x,y) + c(x)\right]$$
$$= \mathbb{E}_{y\sim\pi(\cdot|x)}\left[r(x,y)\right] + c(x)$$

In practice:

$$h_\phi(x,y) \in \mathbb{R}^d$$

$$\Rightarrow r_\phi(x,y) = w^T h_\phi(x,y) + b \in \mathbb{R}$$

$$* \quad r_\phi(x,y) = r^*(x,y) + \varepsilon_\phi(x,y)$$

bias

$$\rightarrow \max_\pi \mathbb{E}_\pi\left[r_\phi(x,y)\right]$$

+ KL reg.

→ "reward hacking"

e.g. length bias.

PPO-based RLHF

$$J_\beta(\theta) = \mathbb{E}_{x\sim D}\left[\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\left[r_\phi(x,y)\right] - \beta \cdot KL\left(\pi_\theta(\cdot|x) \,\|\, \pi_{ref}(\cdot|x)\right)\right]$$

$$\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\left[\log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)}\right]$$

$$= \mathbb{E}_{x,y\sim\pi_\theta}\left[r_\phi(x,y) - \beta \cdot \sum_{t=1}^{T} \frac{\pi_\theta(y_t|s_t)}{\pi_{ref}(y_t|s_t)}\right].$$

"reward shaping": $\tilde{r}_t = \begin{cases} -\beta \cdot k_t, & t < T \\ R - \beta \cdot k_T, & t = T \end{cases}$

$k_t = \log \pi_{\text{old}}(y_t|s_t) - \log \pi_{\text{ref}}(y_t|s_t)$

- value estimation (critic): $V_\psi(s_t) \approx \mathbb{E}_{\pi_{\text{old}}}\left[\sum_{t'=t}^{T} \tilde{r}_{t'} | s_t\right]$
- GAE calculation $\rightarrow \hat{A}_t$
- PPO-clip: define $\rho_t(\theta) = \frac{\pi_\theta(y_t|s_t)}{\pi_{\text{ref}}(y_t|s_t)}$

$$L_k^{\text{PPO}}(\theta) = \mathbb{E}_{\pi_{\text{old}}}\left[\frac{1}{T}\sum_{t=1}^{T} \min\left\{\rho_t(\theta)\cdot\hat{A}_t,\ \text{clip}(\rho_t, 1-\varepsilon, 1+\varepsilon)\cdot\hat{A}_t\right\}\right]$$

"token normalization"

- policy (actor): π_θ ← model
- reference model: π_{ref}. ← π_{old}
- reward model: r_ϕ ← extrapolation error.
- value (critic): V_ψ.

ref_model

reward_model

value_model

DPO (Direct Preference Optimization)

fix prompt x. $J_\beta(\pi) = \mathbb{E}_{y\sim\pi(\cdot|x)}[r(x,y)] - \beta \cdot KL(\pi(\cdot|x) \| \pi_{\text{ref}}(\cdot|x))$

$\max_\pi J_\beta(\pi)$

s.t. $\pi \in \Delta(Y)$ ← π is prob. dist $\pi(\cdot|x) \in \mathbb{R}^{|Y|}$!

$\pi = \pi_\theta$?

$$J_\beta(\pi) = \sum_y \pi(y|x)\left(r(x,y) - \beta \cdot \log \frac{\pi(y|x)}{\pi_{ref}(y|x)}\right)$$

$$\Rightarrow \text{Lagrangian: } \mathcal{L}(\pi, \lambda) = J_\beta(\pi) + \lambda\left(\sum_y \pi(y|x) - 1\right)$$

$$\frac{\partial \mathcal{L}(\pi,\lambda)}{\partial \pi(y|x)} = r(x,y) - \beta\left(\log \frac{\pi(y|x)}{\pi_{ref}(y|x)} + 1\right) + \lambda = 0$$

$$\Rightarrow \log \frac{\pi^*(y|x)}{\pi_{ref}(y|x)} = \frac{r(x,y)}{\beta} - 1 + \frac{\lambda}{\beta}$$

$$\Rightarrow \pi^*(y|x) = \pi_{ref}(y|x) \cdot \exp\left(\frac{r(x,y)}{\beta}\right) \cdot \exp\left(\frac{\lambda}{\beta} - 1\right)$$

$$\pi^*(y|x) = \frac{\pi_{ref}(y|x) \cdot \exp\left(\frac{r(x,y)}{\beta}\right)}{Z(x)}$$

Gibbs policy

normalization: $Z(x) = \sum_y \pi_{ref}(y|x) \cdot \exp\left(\frac{r(x,y)}{\beta}\right)$

"exponential reward tilting"

$$\log \pi^*(y|x) = \log \pi_{ref}(y|x) + \frac{r(x,y)}{\beta} - \log Z(x)$$

$$\Rightarrow r(x,y) = \beta \cdot \log \frac{\pi^*(y|x)}{\pi_{ref}(y|x)} + \beta \cdot \log Z(x)$$

↳ do NOT depend on y

Given a policy π_θ:

↳ $$r_\theta(x,y) = \beta \cdot \log \frac{\pi_\theta(y|x)}{\pi_{ref}(y|x)}$$ ★

$$P(y^+ > y^- | x) = \sigma\left(r(x, y^+) - r(x, y^-)\right)$$

$$= \sigma\left(\beta \cdot \log \frac{\pi_\theta(y^+|x)}{\pi_{ref}(y^+|x)} - \beta \cdot \log \frac{\pi_\theta(y^-|x)}{\pi_{ref}(y^-|x)}\right)$$

⇓ MLE

loss: $\mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(x,y^+,y^-)\sim D_{pref}}\left[\log \sigma\left(\beta \cdot \log \frac{\pi_\theta(y^+|x)}{\pi_{ref}(y^+|x)} - \beta \cdot \log \frac{\pi_\theta(y^-|x)}{\pi_{ref}(y^-|x)}\right)\right]$

Δ_θ

× reward modeling
× critic / GAE
× PPO rollout.

"Preference Optimization"

(x, y^+, y^-): $\ell_{DPO}(\theta) = -\log \sigma(\Delta_\theta)$

$\sigma(z) = \frac{1}{1+e^{-z}}$

$$\nabla_\theta \ell_{DPO}(\theta) = -\frac{\sigma'(\Delta_\theta)}{\sigma(\Delta_\theta)} \cdot \Delta'_\theta$$

$$= -\beta \sigma(-\Delta_\theta)\left(\nabla_\theta \log \pi_\theta(y^+|x) - \nabla_\theta \log \pi_\theta(y^-|x)\right)$$

"supervised learning"

roughly: $\log \pi_\theta(y^+|x) - \log \pi_\theta(y^-|x)$ ↑

SFT: $\mathcal{L}_{SFT}(\theta) = -\mathbb{E}_{(x,y^+)}\left[\log \pi_\theta(y^+|x)\right]$

DPO: $\mathcal{L}_{DPO}(\theta) = -\mathbb{E}_{(x,y^+,y^-)}\left[\log \sigma\left(\beta \cdot \log \frac{\pi_\theta(y^+|x)}{\pi_{ref}(y^+|x)} - \beta \cdot \log \frac{\pi_\theta(y^-|x)}{\pi_{ref}(y^-|x)}\right)\right]$

Reinforcement Learning with Verifiable Rewards (RLVR)

LLM reasoning (LRM)

RLVR

recall: RLHF: human preference → reward model → PPO

↑ reward hacking

↓

$R(x, y) = \text{Verify}(x, y)$

↑

e.g. binary reward. $R(x,y) = \begin{cases} 1 & \text{final answer is correct.} \\ 0 & \text{otherwise} \end{cases}$

1) Math
2) Coding

→ verifier

$(R_1, \cdots, R_G) = (1, 1, \cdots, 1)$

$(R_1, \cdots, R_G) = (0, 0, \cdots, 0)$

3) Theorem proving (lean)
4) Constraint following ....

*In RLVR, not necessary to use reference KL.

data: $D = \{(x_i, a_i^*)\}_{i=1}^n$

→ response $y = (y_1, \cdots, y_T) \sim \pi_\theta(\cdot|x)$

$J_\beta(\theta) = \mathbb{E}[R] - \beta \cdot KL$

→ goal: $\max_\theta J(\theta) = \mathbb{E}_{x \sim D,\, y \sim \pi_\theta(\cdot|x)} [R(x,y)]$

$\left(= \mathbb{P}_{x \sim D,\, y \sim \pi_\theta(\cdot|x)} [\text{answer is correct}] \right)$

$x \longrightarrow y_1, \cdots, y_{T-1} \longrightarrow y_T$

problem — reasoning (?) — final answer → successful

REINFORCE: $\nabla_\theta J(\theta) = \mathbb{E}_{x,y}\left[R(x,y) \cdot \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(y_t|s_t)\right]$

A outcome-based reward (ORM)

↳ long-horizon credit assignment

process-based reward (PRM) ↔ CoT

GRPO (Group Relative Policy Optimization)

critic-free PO.

$$\nabla_\theta J(\theta) = \mathbb{E}_{x,y}\left[\left(R(x,y) - b(x)\right) \cdot \sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(y_t|s_t)\right]$$

$A = Q - V_\phi$

$$b^*(x) = \mathbb{E}_{y \sim \pi_\theta(\cdot|x)}\left[R(x,y)\right]$$

Given prompt x: sample G responses:

$$y^{(1)}, \cdots, y^{(G)} \sim \pi_{\text{old}}(\cdot|x)$$

$$\Rightarrow R_1, \cdots, R_G \quad \text{s.t. } R_i = R(x, y^{(i)})$$

$$\bar{R} = \frac{1}{G}\sum_{i=1}^{G} R_i, \qquad S_R = \text{std}(R_1, \cdots, R_G)$$

binary reward.
$\mathbb{P}[R=1] = P_x$.
$S_R \approx \sqrt{P_x(1-P_x)}$
scaling factor (difficulty)
$\hat{A}_i \approx \frac{R_i - P_x}{\sqrt{P_x(1-P_x)}}$

Advantage estimator: $\hat{A}_i = \frac{R_i - \bar{R}}{S_R + \varepsilon}$

$\Rightarrow \hat{A}_{i,t} = \hat{A}_i, \quad t = 1, 2, \cdots, T_i$

T_i: length of the i-th response

e.g. $(R_1, R_2, R_3, R_4) = (1, 0, 1, 0)$

$\Rightarrow (\hat{A}_1, \hat{A}_2, \hat{A}_3, \hat{A}_4) = (1, -1, 1, -1)$

$$\rho_{i,t}(\theta) = \frac{\pi_\theta(y_t^{(i)} | s_t^{(i)})}{\pi_{\text{old}}(y_t^{(i)} | s_t^{(i)})}$$

$\pi_{\text{old}}(a|s) = 0.01$
$\hat{A} > 0, \ \varepsilon = 0.2$
$\rightarrow 0.012$

$$J_{i,t}(\theta) = \min\left\{ \rho_{i,t}(\theta)\cdot \hat{A}_{i,t}, \ \text{clip}(\rho_{i,t}(\theta), 1-\varepsilon, 1+\varepsilon)\cdot \hat{A}_{i,t} \right\}$$

GRPO objective: $J_{GRPO}(\theta) = \mathbb{E}\left[\frac{1}{G}\sum_{i=1}^{G} \frac{1}{T_i}\sum_{t=1}^{T_i} J_{i,t}(\theta) \right]$

per-response-normalized

$$\hat{g} = \frac{1}{G}\sum_{i=1}^{G} (R_i - \bar{R}) \cdot \nabla_\theta \log \pi_\theta(y^{(i)} | x)$$

$\Rightarrow \mathbb{E}[\hat{g} | x] = \left(1 - \frac{1}{G}\right) \cdot \nabla_\theta J_x(\theta)$

* Leave-One-Out (LOO): $b_{-i} = \frac{1}{G-1}\sum_{j \neq i} R_j$.

$J(\theta) = \mathbb{E}_{x \sim D}[J_x(\theta)]$.

Variants of GRPO

- **DAPO**
  1) Clip-Higher: $\text{clip}(\rho, 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}})$. $\varepsilon_{\text{high}} > \varepsilon_{\text{low}}$
  2) Dynamic sampling: $0 < \sum_{i=1}^{G} R_i < G$. → curriculum learning. (evolve difficulty filtering)
  3) Token-level policy optimization:
  $$\mathcal{L}_{\text{DAPO}} = \frac{1}{\sum_{i=1}^{G} T_i} \sum_{i=1}^{G} \sum_{t=1}^{T_i} l_{i,t}$$
  4) Overlong reward shaping.
     - overlong filtering.
     - soft overlong punishment.

- **Dr. GRPO**:
$$\mathcal{L}_{\text{Dr.GRPO}}(\theta) = \frac{1}{G \cdot L_{\max}} \sum_{i=1}^{G} \sum_{t=1}^{T_i} l_{i,t}. \qquad \hat{A}_i = R_i - \bar{R}.$$
$L_{\max}$: fixed normalization constant.

- **GSPO**:
length-normalized sequence ratio. $S_i(\theta) = \left(\frac{\pi_\theta(y^{(i)}|x)}{\pi_{\theta_{\text{old}}}(y^{(i)}|x)}\right)^{1/T_i}$
$$= \exp\left(\frac{1}{T_i}\sum_{t=1}^{T_i} \frac{\pi_\theta(y_t^{(i)}|s_t^{(i)})}{\pi_{\theta_{\text{old}}}(y_t^{(i)}|s_t^{(i)})}\right)$$
$$\Rightarrow \mathcal{L}_{\text{GSPO}}(\theta) = \frac{1}{G}\sum_{i=1}^{G} \min\{S_i(\theta)\hat{A}_i, \text{clip} \cdots \hat{A}_i\}$$
.....

Challenges

1) verifier. String matching

\frac12

$\frac{1}{2} = 0.5 = \frac{2}{4}$

→ Symbolic equivalence checking.

2) exploration & sparse reward: middle-difficulty problems

$P_x \rightarrow 1-(1-P_x)^G - P_x^G$.

3) Credit assignment: $\hat{A}_{i,t} = \hat{A}_i, \ \forall t \in [T_i]$.

$Q^\pi(s_t, a_t) = P_\pi[\text{eventually correct} \mid s_t, a_t]$.

4) Long-CoT → asynchronous, traing.

5) entropy collapse.

$$H(\pi_\theta(\cdot|s)) = -\sum_a \pi_\theta(a|s) \cdot \log \pi_\theta(a|s).$$

△ RLVR
- Sharpener: reweighting (sample effiviency ↑)
- Discoverer: new reasoning paradigms.

limit-of-RLVR

pass@k
- k small: RL > base.
- k large: RL < base.

Regime ☆

base model → exploration (AlphaGo #37 Move)

$f(g(x))$

On-Policy Distillation (OPD)

knowledge distillation: teacher: 32B; student: 4B → trasfer the reasoning capability from teacher to student.

- off-policy distillation: $x \sim D$. $y \sim q(\cdot|x)$ q: teacher model,

$$\max_\theta \mathbb{E}_{x\sim D,\ y\sim q(\cdot|x)}\left[\log \pi_\theta(y|x)\right] \quad \text{(SFT)}$$

$\Leftrightarrow$ minimize $\mathbb{E}_{x\sim D}\left[KL\left(q(\cdot|x) \,\|\, \pi_\theta(\cdot|x)\right)\right]$

distribution shift

teacher || student → forward KL

(mode covering)

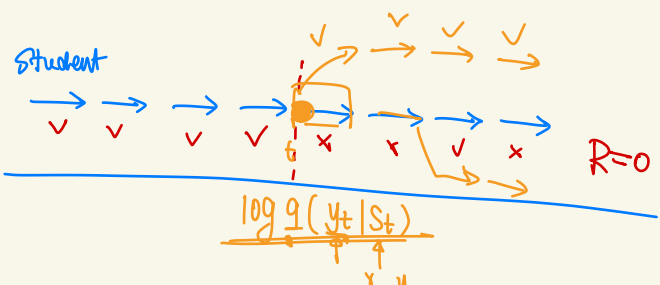
- RLVR → sparse reward.

OPD
- SFT: supervision
- RLVR: on-policy

↓

dense (token-level) feedback,

+ on-policy sampling



Setup.

① minimize reverse KL: $\mathbb{E}_{x\sim D}\left[KL\left(\pi_\theta(\cdot|x) \,\|\, q(\cdot|x)\right)\right]$ (mode seeking; Student || teacher) → q: teacher policy

$$\downarrow$$

$$\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}\left[\log \frac{\pi_\theta(y|x)}{q(y|x)}\right]$$

$$\max_\theta \sum_{t=1}^{T} \mathbb{E}_{y_t\sim\pi_\theta(\cdot|s_t)}\left[\log \frac{q(y_t|s_t)}{\pi_\theta(y_t|s_t)}\right]$$

$$r_t^{OPD} = \log q(y_t|s_t) - \log \pi_\theta(y_t|s_t)$$

"reward" ×

$q(y_t|s_t) > \pi_\theta(y_t|s_t)$: $r_t^{OPD} > 0$,

$q(y_t|s_t) < \pi_\theta(y_t|s_t)$: $r_t^{OPD} < 0$.

② DPO: $\max_\pi \mathbb{E}_\pi[r(x,y)] - \beta \cdot KL(\pi \| \pi_{ref})$

$\Rightarrow \pi^*(y|x) \propto \pi_{ref}(y|x) \cdot \exp\left(\frac{r(x,y)}{\beta}\right)$

OPD: q

$\Rightarrow r_q(x,y) = \beta \cdot \log \frac{q(y|x)}{\pi_{ref}(y|x)}$ "reward" ✓

$\Rightarrow \mathbb{E}_{y\sim\pi(\cdot|x)}[r_q(x,y)] - \beta \cdot KL(\pi \| \pi_{ref})$ KL-regularized obj.

$= \beta \cdot \mathbb{E}_{y\sim\pi(\cdot|x)}\left[\log \frac{q(y|x)}{\pi_{ref}(y|x)} - \log \frac{\pi(y|x)}{\pi_{ref}(y|x)}\right]$

$= \beta \cdot \mathbb{E}_{y\sim\pi(\cdot|x)}\left[\log \frac{q(y|x)}{\pi(y|x)}\right]$

$= -\beta \cdot KL(\pi(\cdot|x) \| q(\cdot|x))$

③ $\max_\theta J_{OPD}(\theta) := \mathbb{E}_{x,y\sim\pi_\theta(\cdot|x)}[\log q(y|x) - \log \pi_\theta(y|x)]$

$= \mathbb{E}_x\left[\mathbb{E}_{y\sim\pi_\theta(\cdot|x)}[\log q(y|x)] + \mathcal{H}(\pi_\theta(\cdot|x))\right]$

↑ entropy-regularized RL

$\Rightarrow \pi^*(\cdot|x) = q(\cdot|x)$.

Policy Gradient

$r_\theta(y) = \log q(y|x) - \log \pi_\theta(y|x)$

$\nabla_\theta J_{OPD}(\theta) = \nabla_\theta \mathbb{E}_{\pi_\theta}[r_\theta(y)]$

$$= \mathbb{E}_{\pi_\theta}\left[r_\theta(y)\, \nabla_\theta \log \pi_\theta(y|x) + \nabla_\theta r_\theta(y) \right]$$

$$= \mathbb{E}_{\pi_\theta}\left[r_\theta(y) \cdot \nabla_\theta \log \pi_\theta(y|x) - \nabla_\theta \log \pi_\theta(y|x) \right]$$

$$= \mathbb{E}_{\pi_\theta}\left[r_\theta(y) \cdot \nabla_\theta \log \pi_\theta(y|x) \right]$$

$\mathbb{E}_{x\sim p}[\nabla \log p(x)]$
$= \sum_x p(x) \cdot \frac{\nabla p(x)}{p(x)}$
$= \sum_x \nabla p(x)$
$= \nabla \sum_x p(x) = 0$

★ $$\nabla_\theta J_{OPD}(\theta) = \mathbb{E}_{x,y\sim\pi_\theta(\cdot|x)}\left[\left(\log q(y|x) - \log \pi_\theta(y|x) \right) \cdot \nabla_\theta \log \pi_\theta(y|x) \right]$$

$$= \mathbb{E}\left[\left(\sum_{t=1}^{T} r_t^{OPD} \right) \left(\sum_{t=1}^{T} \nabla_\theta \log \pi_\theta(y_t|s_t) \right) \right]$$

↓ Causality

$$= \mathbb{E}\left[\sum_{t=1}^{T} \left(\sum_{t'=t}^{T} r_{t'}^{OPD} \right) \cdot \nabla_\theta \log \pi_\theta(y_t|s_t) \right]$$

G_t^{OPD} : reward-to-go.

① current token.

② future tokens

In practice:

$$g_{local} = \mathbb{E}\left[\sum_{t=1}^{T} r_t^{OPD} \cdot \nabla_\theta \log \pi_\theta(y_t|s_t) \right].$$

↳ fix prefix s:

$$L_s(\theta) = D_{KL}\left(\pi_\theta(\cdot|s) \,\|\, q(\cdot|s) \right),$$

$$= \sum_a \pi_\theta(a|s) \cdot \log \frac{\pi_\theta(a|s)}{q(a|s)}$$

$$\nabla_\theta L_s(\theta) = \sum_a \nabla_\theta \pi_\theta(a|s) \cdot \left(\log \frac{\pi_\theta(a|s)}{q(a|s)} + 1 \right)$$

$$= \mathbb{E}_{a\sim\pi_\theta(\cdot|s)}\left[\log \frac{\pi_\theta(a|s)}{q(a|s)} \cdot \nabla_\theta \log \pi_\theta(a|s) \right]$$

$\log \frac{\pi_\theta(a|s)}{q(a|s)} = r^{OPD}(s,a)$

1. sample prompts

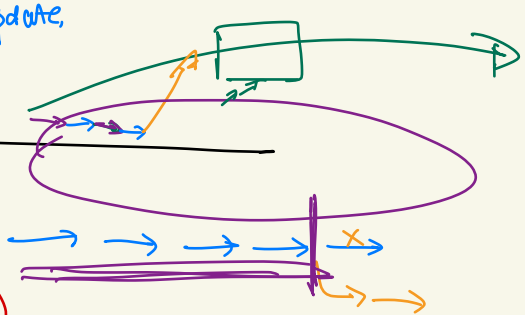
2. student rollout ("off-policy") behavior policy $y \sim \pi_{\text{old}}(\cdot|x)$

3. teacher scoring $\log q(y_t|s_t)$

4. construct reward $r_t^{\text{OPD}} = \log q(y_t|s_t) - \log \pi_{\theta}(y_t|s_t)$ π_{old}

5. PG-style update / PPO-style update

6. refresh student rollouts



OPD vs RL

exploration

$y \sim \pi_\theta(\cdot|x)$

(local) correction

4B

32B

$s_t \sim d_t^{\pi_\theta}$ → student.

SFT: $s_t \sim d_t^{q}$ → teacher

distribution mismatch

SFT/off-policy

↓

OPD ← coverage ?

function class

data reuse.

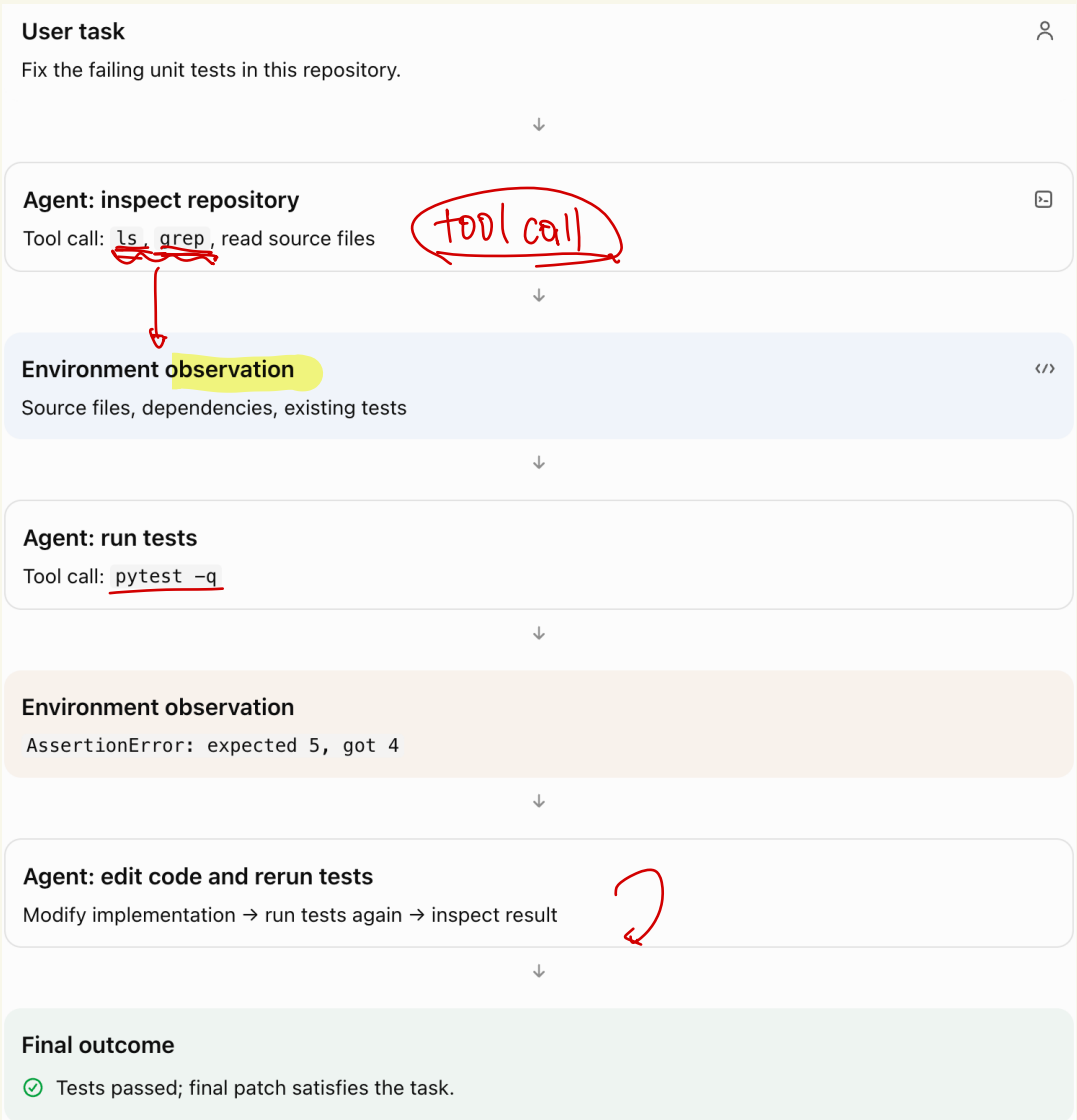
Thinking Machine.

	trajectory from	supervision / feedback
OPD	student	dense teacher scoring
SFT	teacher	demonstration
RL	student	sparse outcome reward

Agentic Reinforcement Learning

LLM: question-answer

Coding Agent: interaction



RLVR: $x \to y \to R(x,y)$

Agentic RL: $x \to a_1 \to o_2 \to a_2 \to o_3 \to \cdots \to R$.

a: action
o: observation.

- tool-augmented LLM.
- LLM agent: ReAct.
- Agentic RL: reward → update agent.

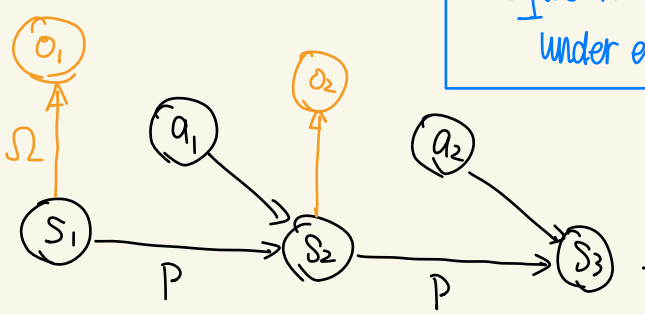
POMDP (Partial Observable Markov Decision Process)

$(S, A, O, P, \Omega, R, \gamma, d_0)$

Agent-Enviroment Interaction + Policy Learning from Reward,

↓

Sequential decision making under environmental feedback.



MDP: $s_k = o_k$.

$(\Omega = I)$

State: s

observation: o → history $h_k = (o_1, a_1, o_2, a_2, \cdots, o_k)$

history-dependent policy: $a_k \sim \pi_\theta(\cdot | h_k)$

(latent) state transition: $s_{k+1} \sim P(\cdot | s_k, a_k)$

emission: $o_{k+1} \sim \Omega(\cdot | s_{k+1})$

(Outer-loop)

High-level: Agent decision steps.

e.g. Read files → Run tests → Edit code.

Low-level: LLM token generation.

(inner-loop)

e.g. {"tool": "bash", "command": "pytest -q"}

$$a_k = (\text{tool name}, \text{arguments})$$

$$\pi_\theta(a_k \mid h_k) = \prod_{t=1}^{T_k} \pi_\theta(a_{k,t} \mid h_k, a_{k,<t})$$

trajectory $\tau = (s_1, o_1, a_1, s_2, o_2, a_2, \cdots, s_{k+1}, o_{k+1}) \rightarrow R(\tau)$

$$\Rightarrow P_\theta(\tau) = d_0(s_1) \cdot \Omega(o_1 \mid s_1) \cdot \prod_{k=1}^{k} \pi_\theta(a_k \mid h_k) \cdot P(s_{k+1} \mid s_k, a_k) \cdot \Omega(o_{k+1} \mid s_{k+1})$$

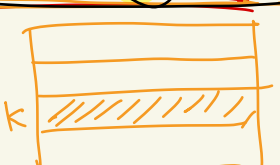
condition

$$\Rightarrow \nabla_\theta \log P_\theta(\tau) = \sum_{k=1}^{k} \nabla_\theta \log \pi_\theta(a_k \mid h_k) \quad \checkmark$$

$$\nabla_\theta J(\theta) = \mathbb{E}_{\tau \sim P_\theta(\tau)} \left[R(\tau) \cdot \sum_{k=1}^{k} \nabla_\theta \log \pi_\theta(a_k \mid h_k) \right]$$

$$= \mathbb{E}_{\tau \sim P_\theta(\tau)} \left[R(\tau) \cdot \sum_{k=1}^{k} \sum_{t=1}^{T_k} \nabla_\theta \log \pi_\theta(a_{k,t} \mid h_k, a_{k,<t}) \right]$$

- ✓ agent generation → $a_k \sim \pi_\theta(\cdot \mid h_k)$
- ✗ tool observation
- ✓ agent generation



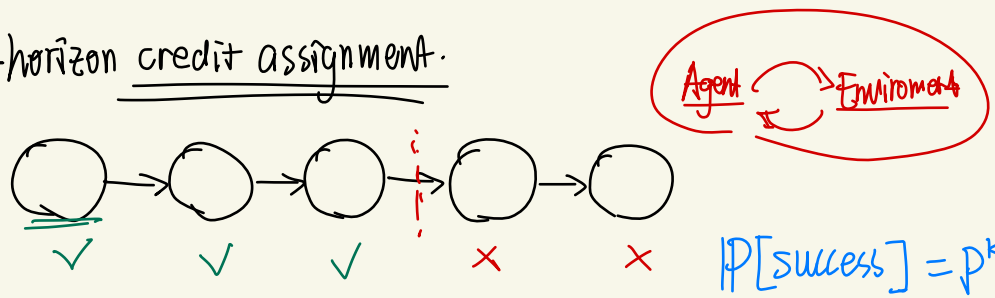
token sequence.

$$L(\theta) = \sum_{j=1}^{N} m_j A_j \log \pi_\theta(z_j \mid z_{<j})$$

★ mask $m_j = \begin{cases} 1, & z_j \text{ is generated by trained policy} \\ 0, & z_j \text{ is outside token from prompt / tool observation} \end{cases}$

rollout → multi-turn RL

Long-horizon credit assignment.



$P[\text{success}] = p^k$.

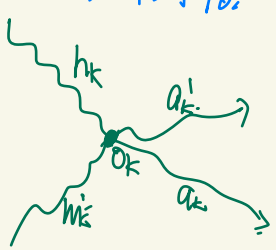
$p=0.95$
$k=50$
$\Rightarrow 7.7\%$.

1) Turn-level value function

$V^\pi(s_k)$
$Q^\pi(s_k, a_k)$

→ $V^\pi(o_k)$
$Q^\pi(o_k, a_k)$ ×

$V^\pi(h_k)$
$Q^\pi(h_k, a_k)$



$h_k = (o_1, a_1, o_2, a_2, \cdots, o_k) \in H_k$.

$A^\pi(h_k, a_k) = Q^\pi(h_k, a_k) - V^\pi(h_k)$

$|H_k| = |O|^k \cdot |A|^{k-1}$.

↳ exponential (curse of horizon)

$$\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{k=1}^{k} \gamma^{k-1} \cdot \nabla_\theta \log \pi_\theta(a_k \mid h_k) \cdot A^\pi(h_k, a_k)\right]$$

2) Process Reward / Intermediate Feedback.

↓ potential-based reward shaping (e.g. GTPO)

↓ Change the optimization objective.

3) Tree search / Counterfactual rollouts.

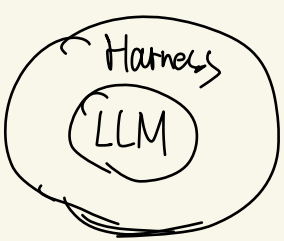
$h_k \rightarrow \begin{cases} a_k^{(1)} \rightarrow R^{(1)} \\ a_k^{(2)} \rightarrow R^{(2)} \\ a_k^{(3)} \rightarrow R^{(3)} \end{cases} \rightarrow Q^{\pi}(h_k, a)$

reset ?

MLSys

Agent = LLM policy + Harness + Environment

(execution framework)



Task → LLM policy → control logic → environment → reward / evaluation

* Trainer → LLM → tool → trainer

training-deployment mismatch

1) environment reset.

2) multi-turn rollout.

3) reward evaluation

4) trajectory processing.

5) advantage estimation.

6) policy optimization

7) held-out evaluation

2 kinds of resources:

- GPU resources
- environment resources.

↓

Asynchronous rollout!

(parallel)

↓

off-policyness

$$\rho(\tau) = \prod_{k=1}^{k} \frac{\pi_\theta(a_k|h_k)}{\mu(a_k|h_k)}$$

Enviroment ⟺ Agent

e.g. coding agent OpenAI-HuggingFace

human supervision ← AI safety, AI alignment.

PG: REINFORCE. baseline. GAE. PPO. TRPO. NPG.

↓

LLM-RL: RLHF, RLVR, Agentic RL.